

Towards a Complete Formalization of PLN (DRAFT)

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1 Introduction

The goal is similar to Solomonoff Universal Induction [?], that is we want to approach a first order (unknown but computable) distribution μ given observations, using a second order (uncomputable but known) distribution ν , called the Universal Distribution. In Solomonoff Induction, observations are bit strings produced by a Turing machine¹. In PLN however, observations are outcomes from an indexed boolean random variable, representing the outputs of evaluating a predicate on some inputs. Such predicate is called the *observable predicate*. In practice PLN allows multiple observable predicates however one can assume one predicate without loss of generality. Indeed, to emulate multiple predicates, one can introduce an extra component in the predicate's domain to “select” the predicate of interest. Also, since it is observed by a random variable, such predicate is not necessarily deterministic (though it could be). As such, one may think of the observable predicate as being a program drawn from a certain Probabilistic Programming Language. In the following section we formally define the above.

Because this reformulation of PLN departs somewhat from the definition of PLN in the PLN book [?], we give it a new name, ν PLN.

2 Definitions

In its original form, PLN purposely avoids relying on an underlying global probability distribution. I am not against this in principle. I will simply admit that I cannot conceive a complete definition of PLN that does not rely on such global probability distribution. I would also point out that a publication released after the PLN book by the principal authors of the PLN book, Ben Goertzel and Matt Ikle, very much aligns with the idea of a global probability distribution [?], and was in fact a great source of inspiration for writing this very document. The

¹Note that even though the sample space of ν is made of deterministic Turing machines, ν can approximate any computable distribution μ (thus non-deterministic) by maintaining an ensemble of such Turing machines.

next subsection is dedicated to define the global probability distribution which ν PLN is intended to derive from.

2.1 Global Probability Distribution

Let $(\Omega, \mathcal{F}, \nu)$ be a probability space such that

- $\mathcal{F}$ is the event space, a σ -algebra on Ω .
- $\nu : \mathcal{F} \rightarrow [0, 1]$ is a universal distribution, further defined below.
- Ω , the set of possible worlds, is the sample space associated to ν , such that each element $\hat{\omega} \in \Omega$ contains
 1. a probabilistic predicate $\hat{\mu} \in \mathcal{L}$ over a domain $\mathcal{D}$, described in a probabilistic programming language $\mathcal{L}$,
 2. a mapping of $\hat{\mu}$ from $\mathcal{D}$ to Boolean. For instance if $\mathcal{D}$ is $\mathbb{N}$, then a possible mapping could be $(\hat{\mu} \ 0) = \text{True}$, $(\hat{\mu} \ 1) = \text{False}$, $(\hat{\mu} \ 2) = \text{True}, \dots$, corresponding so far to a predicate indicating the evenness of a natural number. Note that the world $\hat{\omega}$ contains the entire, potentially infinite, mapping, even though in reality an observer can only have access to a finite subset of it.

An observer lives in a particular world $\omega \in \Omega$, called the *true world*, which includes the *true generator* or *true predicate*, μ , and the *true history*, which is the complete mapping of evaluations of μ over the domain $\mathcal{D}$. Note that since μ is probabilistic, it does not deterministically determine the history, thus the true history is just one possible history among an infinity of histories compatible with μ . Note that throughout the document ω and μ refer to the true world and true generator respectively, which is to be contrasted with $\hat{\omega}$ and $\hat{\mu}$ which refer to arbitrary elements of Ω and $\mathcal{L}$ respectively. Although when it is clear that the definitions are generic and apply to any arbitrary world and the true world alike, we may use ω and μ as well. Since μ is a probabilistic predicate, its type signature cannot merely be

$$\mu : \mathcal{D} \rightarrow \text{Bool}$$

where $\text{Bool} = \{\text{False}, \text{True}\}$. To capture its probabilistic nature we give it the following type signature

$$\mu : \mathcal{D} \rightarrow \Omega \rightarrow \text{Bool}$$

in a curried fashion. Meaning that given its argument, it produces a boolean random variable. Therefore the observable predicate can be viewed as an indexed boolean random variable. Of course, μ never gets to be evaluated on a different world than the one it belongs to, thus we can write $(\mu \ a)$ while meaning $(\mu \ a \ \omega)$, but we still need to keep the Ω argument in order to reason about possible worlds since the true world is unknown. Also, in cases where the domain

$\mathcal{D}$ can be decomposed into multiple components, a curried notation will be used interchangeably. So instead of

$$\mu : (\mathcal{D}_1 \times \dots \times \mathcal{D}_n) \rightarrow \Omega \rightarrow \text{Bool}$$

the following

$$\mu : \mathcal{D}_1 \rightarrow \dots \rightarrow \mathcal{D}_n \rightarrow \Omega \rightarrow \text{Bool}$$

will be used. This will be convenient to express inheritance relationships between partially applied predicates. As already hinted, the application of μ to an input x is denoted with the traditional functional programming style

$$(\mu \ x)$$

Thus if the domain is decomposed into subdomains $\mathcal{D}_1$ to $\mathcal{D}_n$, applying μ to all its inputs x_1 to x_n will be denoted

$$(\mu \ x_1 \ \dots \ x_n)$$

Likewise for the ω argument

$$(\mu \ x_1 \ \dots \ x_n \ \omega)$$

Such functional programming notation is used for μ because it is currying-friendly. For the rest, we keep using the traditional mathematical function application style, such as

$$\nu(E)$$

denoting the application of the probability distribution ν to the event E . In case μ is known to be deterministic, Ω could potentially be dropped, but that is not going to be our working assumption for now. Additionally to the functional programming style, we may use parametric notation, so for instance instead of

$$(\mu \ x_1 \ x_2)$$

we may write

$$(\mu_{x_1} \ x_2)$$

where x_1 is viewed as a parameter and x_2 is viewed as the argument of μ_{x_1} . With that, let us now define key random variables to access Ω :

- $M : \Omega \rightarrow \mathcal{L}$ with measurable space $(\mathcal{L}, \mathcal{F}_{\mathcal{L}})$, where $\mathcal{L}$ is a certain probabilistic programming language and $\mathcal{F}_{\mathcal{L}}$ is a σ -algebra on $\mathcal{L}$. Thus, M takes a world $\hat{\omega} \in \Omega$ and outputs the probabilistic program $\hat{\mu} \in \mathcal{L}$ generating that world. Note that this random variable is inaccessible from an observer within that world. An observer within that world only has access to a finite record of evaluations of $\hat{\mu}$. However, this random variable is important to reason about multiple worlds, we suspect it is particularly important for higher order reasoning.

- $D_{x \in \mathcal{D}} : \Omega \rightarrow \text{Bool}$, a Boolean random variable indexed by values in $\mathcal{D}$. Unlike M , $D_{x \in \mathcal{D}}$ is at least partially accessible from an observer within that world. Meaning, such observer can gather data for a finite subset $\mathcal{S}$ of $\mathcal{D}$. In this case $D_{\mathcal{S}}$ represents a finite family of Boolean random variables, corresponding the set of accessible observations. M and $D_{x \in \mathcal{D}}$ are related by the following equality

$$(\hat{\mu} \ x \ \hat{\omega}) = (D_x \ \hat{\omega})$$

where $\hat{\omega} \in \Omega$ such that $(M \ \hat{\omega}) = \hat{\mu}$. Or simply, in curried fashion

$$(\hat{\mu} \ x) = D_x$$

Due to the equality above, the distribution of observations is entirely determined by a model $\hat{\mu}$. In other words, it suffices to define a distribution over $\mathcal{L}$, the prior distribution over possible models, to define ν (as far as M and D_x are concerned anyway). Then, relating observations to models can be done using regular Bayesian inference. The prior is defined by

$$\nu(M \in L)$$

where $L \in \mathcal{F}_{\mathcal{L}}$. Note how it is expressed in terms of elements of $\mathcal{F}_{\mathcal{L}}$, instead of elements of $\mathcal{L}$. It is because, for the purpose of recovering PLN with the Bayesian approach, $\mathcal{L}$ needs to be continuous. I will explain this in detail, but for now let us use this notation to formulate Bayes' theorem

$$\nu(M \in L | D_{\mathcal{S}}) = \frac{\nu(M \in L) \times \nu(D_{\mathcal{S}} | M \in L)}{\nu(D_{\mathcal{S}})} \quad (1)$$

where $\mathcal{S}$ is a finite subset of $\mathcal{D}$. Sometimes, for the sake of simplicity, we may drop M since it is the only random variable that ranges over $\mathcal{L}$, and write instead

$$\nu(L | D_{\mathcal{S}}) = \frac{\nu(L) \times \nu(D_{\mathcal{S}} | L)}{\nu(D_{\mathcal{S}})} \quad (2)$$

An example of prior will be given in 4, but in general it can be viewed as a parameter of νPLN . That is, given a certain prior of ν over $\mathcal{L}$, one can derive a certain flavor of νPLN . Since $(\mu \ x) = D_x$, a history data point will be represented as $(\mu \ x) = \text{True}$ or $(\mu \ x) = \text{False}$, and the usage of the random variable D_x , and especially $D_{\mathcal{S}}$, will be reserved in the formulation of ν to refer to the observed history.

2.2 PLN Global Distribution vs Solomonoff Induction

Note that unlike with Solomonoff Universal Induction, $\hat{\mu}$ represents a probabilistic predicate rather than a computable probability function calculating the probability of any event, although the latter can be derived from the former.

The *true distribution* in Solomonoff Induction corresponds to the *real predicate* here. We prefer to use the word *real* when denoting objective reality, rather than *true* because *true predicate* could be understood as a predicate that always outputs true.

3 PLN and the Global Distribution

In Section 4 will proceed to derive PLN rules from the global probability distribution introduced in Section 2.1, but for now let us recall important notions of PLN and how to relate them to the global distribution defined in Section 2.1.

3.1 Concepts vs Predicates

The PLN book describes respectively the notions of *concepts* and *predicates*, and their respective associated relationships *inheritance* and *implication*. As explain in Section 2.6 *Higher-Order Logical Relationships* of the PLN book, there is a perfect isomorphism between concepts and inheritance on one side and predicates and implication on the other side. Concerned with conciseness, we will pick a side, the predicate side, and essentially forget about the other side, the concept side, for the rest of this document. But before we do so let us recall what is a concept, the inheritance between two concepts, and the isomorphism between concepts and predicates.

3.1.1 Concepts and Inheritance

A concept is a fuzzy (or, as I prefer to say, probabilistic, for reasons I will explain in Section ??) set, and the *extensional* inheritance between two concepts is a probabilitized subset relationship. Originally, in the PLN book, the inheritance relationship is defined as an explicit mixture of extensional and intensional inheritances. We will show however that they are in fact both the same thing, the extensional inheritance is a way to approach inheritance solely via *induction*, and intensional inheritance is a way to approach inheritance solely via *abduction*. Any one side, extensional or intensional, is good enough to define the other, so let us pick one, the extensional side, and define inheritance with it. The extensional inheritance between two concepts can be viewed as a probabilitized subset relationship. It allows to express things like “most members of a set are also members of another set”. For instance one could express in PLN that 90% of birds fly, with

$$(\text{Inheritance Bird Fly}) \stackrel{\text{m}}{=} 0.9$$

Such knowledge might have been obtained by observing a finite sample of birds and whether or not they fly. Meaning there could be an uncertainty on the 90% itself. To represent such uncertainty PLN uses a second order distribution, in this case a beta distribution as it is an ideal choice to represent the posterior of the parameter of a Bernoulli distribution given observations. Under this assumption only two numbers are required to determine the parameters, α and β , of the associated beta distribution. A *truth value* called *simple truth value* was created for this purpose and is thus described by two numbers: a strength (a proxy for a probability estimate) and a confidence over this strength from which the α and β parameters of the beta distribution can be recovered. For

instance, given a simple truth value one may express that 90% of birds fly with a confidence of 0.99

$$(\text{Inheritance Bird Fly}) \stackrel{\text{m}}{=} \langle 0.9, 0.99 \rangle$$

where 0.9 is the strength and 0.99 is the confidence. The confidence is a value between 0 and 1 that actually encodes the sample size that was used to obtain the strength. The higher the sample size, the higher the confidence. More information about that will be provided in Section ?? but for now let us just leave it at that as it is enough for what we are concerned with in this Section which is the isomorphism between concepts and predicates.

3.1.2 Isomorphism between Concepts and Predicates

To every concept one can associate a predicate and vice versa. To go from concept to predicate one can use the *indicator function*, and to go from predicate to concept one can use the *satisfying set*. These notions are well known for crisp predicates and sets and thus will not be detailed any further here. The only difference in PLN is that concepts are probabilistic, meaning that a probability (or potentially a second order probability) can be attached to the membership of an element to a concept. Likewise, predicates are probabilistic in the sense that a (second order) probability can be attached to the evaluation of an argument to a predicate. As one would expect the isomorphism also applies between the inheritance relationship on the concept side and the implication relationship on the predicate side. So for instance, on the predicate side one can express the same inheritance relationship between birds and fly as follows

$$(\text{Implication IsBird DoesFly}) \stackrel{\text{m}}{=} \langle 0.9, 0.99 \rangle$$

where IsBird and DoesFly have been obtained by taking the indicator functions of Bird and Fly. As mentioned earlier, we will drop the concept side and only focus on the predicate side for the remaining of the document.

3.2 Relating Predicates to μ

As explained in Section 2.1 there is only one global predicate, μ . To manipulate multiple predicates we can simply introduce a additional parameter in the domain of μ indicating predicates. Let us call this subdomain $\mathcal{P}$ which ranges over symbols representing predicates, in this case μ may have the type signature

$$\mu : \mathcal{P} \rightarrow \mathcal{D} \rightarrow \Omega \rightarrow \text{Bool}$$

So for instance to express that a cat is a mammal, traditionally represented as

$$Cat \Rightarrow Mammal$$

one may use the following parametric notation of μ

$$\mu_{Cat} \Rightarrow \mu_{Mammal}$$

Likewise, to represent the evaluation of the predicate *Cat* over a certain cat instance, cat_{123} , one may use

$$(\mu_{Cat} \ cat_{123})$$

corresponding to the traditional representation

$$(Cat \ cat_{123})$$

where $cat_{123} \in \mathcal{D}$.

3.3 Truth Value as Posterior Distribution

3.4 Statement versus Judgement

Like in NAL, a PLN *statement* designates a logical statement without truth value, such as

$$P \Rightarrow Q$$

While a PLN *judgment* designates a PLN statement with a truth value attached to it, such as

$$P \Rightarrow Q \stackrel{m}{=} \langle 0.9, 0.8 \rangle$$

4 Deriving PLN Rules

Our goal here is to derive every PLN rules in the PLN book from the global distribution that has been defined in Section 2.1. By doing so we hope not only to provide a clear unambiguous definition for each rule, but also an ideal to approach as using a global distribution should give us the means to derive a convergence theorem a la Solomonoff.

4.1 Predicate Direct Introduction

This rule is meant to calculate the truth value attached to a predicate using direct observations. Its truth value can be understood as the marginal probability of a predicate to be true, irrespective of the inputs. It can be viewed as a subcase of the Implication Direct Introduction presented in Section 4.2 where the implicant is the *Universal Predicate* (a predicate that is always true), but is described here as its own rule to simplify the presentation. Indeed, it should be easier to understand the Implication Direct Introduction rule after understanding the Predicate Direct Introduction rule.

To derive the predicate direct introduction rule we consider the posterior probability of a Bernoulli process with parameter p given all available observations of the predicate in question. Let us begin with μ itself, the overarching predicate. In order to formulate this posterior we need to assume that $\mathcal{L}$ has a Bernoulli sampler, or Bernoulli process, in its set of operations. Let $\mathcal{B}$ be the subset of programs of $\mathcal{L}$ consisting of a single Bernoulli call, thus $\mathcal{B}$ is comprised of the following programs

$$(\text{Bernoulli } \theta)$$

where θ is the parameter of the Bernoulli distribution ranging over $[0, 1]$. Such probabilistic program, when executed, outputs **True** with a probability of θ or **False** with a probability of $1 - \theta$. Let $\mathcal{F}_{\mathcal{B}}$ be a Borel σ -algebra on $\mathcal{B}$. We can assume that $\mathcal{F}_{\mathcal{B}}$ is a subfield of $\mathcal{F}_{\mathcal{L}}$. Let us first derive a set of generic equations involving the event $L_{\mathcal{B}} \in \mathcal{F}_{\mathcal{B}}$ knowing observations $D_{\mathcal{S}}$ and the fact that the probabilistic programs under consideration are limited to Bernoulli processes, that is $M \in \mathcal{B}$:

$$\nu(M \in L_{\mathcal{B}} | D_{\mathcal{S}}, M \in \mathcal{B})$$

$$= \frac{\nu(M \in L_{\mathcal{B}}, D_{\mathcal{S}}, M \in \mathcal{B})}{\nu(D_{\mathcal{S}}, M \in \mathcal{B})} \quad (3)$$

$$= \frac{\nu(D_{\mathcal{S}} | M \in L_{\mathcal{B}}, M \in \mathcal{B}) \times \nu(M \in L_{\mathcal{B}}, M \in \mathcal{B})}{\nu(D_{\mathcal{S}}, M \in \mathcal{B})} \quad (4)$$

$$= \frac{\nu(D_{\mathcal{S}} | M \in L_{\mathcal{B}}) \times \nu(M \in L_{\mathcal{B}}, M \in \mathcal{B})}{\nu(D_{\mathcal{S}}, M \in \mathcal{B})} \quad (5)$$

$$= \frac{\nu(D_{\mathcal{S}} | M \in L_{\mathcal{B}}) \times \nu(M \in L_{\mathcal{B}} | M \in \mathcal{B}) \times \nu(M \in \mathcal{B})}{\nu(D_{\mathcal{S}} | M \in \mathcal{B}) \times \nu(M \in \mathcal{B})} \quad (6)$$

$$= \frac{\nu(D_{\mathcal{S}} | M \in L_{\mathcal{B}}) \times \nu(M \in L_{\mathcal{B}} | M \in \mathcal{B})}{\nu(D_{\mathcal{S}} | M \in \mathcal{B})} \quad (7)$$

$$= \frac{\prod_{i \in \mathcal{S}} \nu(D_i | M \in L_{\mathcal{B}}) \times \nu(M \in L_{\mathcal{B}} | M \in \mathcal{B})}{\prod_{i \in \mathcal{S}} \nu(D_i | M \in \mathcal{B})} \quad (8)$$

Let us walk through each equation one by one. Equation 3 is derived from the definition of conditional probability, so is Equation 4. The latter is a specialization of Equation 1 restricted to Bernoulli processes. Equation 5 is obtained from the fact that $L_{\mathcal{B}} \subseteq \mathcal{B}$. Equation 6 is derived from the definition of conditional probability. Equation 7 is obtained by canceling out $\nu(M \in \mathcal{B})$ in the numerator and denominator. If $M \in \mathcal{B}$ then observations are independent and identically distributed (iid), therefore $\nu(D_{\mathcal{S}} | M \in \mathcal{B}) = \prod_{i \in \mathcal{S}} \nu(D_i | M \in L_{\mathcal{B}})$. The same applies to $\nu(D_{\mathcal{S}} | M \in L_{\mathcal{B}})$ since $L_{\mathcal{B}} \subseteq \mathcal{B}$, resulting in Equation 8.

We claim that the PLN truth value of μ , $\text{TV}(\mu)$, expressed as the cumulative distribution function of its corresponding second order probability distribution, $\text{CDF}_{\text{TV}(\mu)}$, is defined as follows

$$\text{CDF}_{\text{TV}(\mu)}(x) = \nu(M \in L_{\mathcal{B}[0,x]} | D_{\mathcal{S}}, M \in \mathcal{B}) \quad (9)$$

where $L_{\mathcal{B}[0,x]} \in \mathcal{F}_{\mathcal{B}}$ is the set of Bernoulli processes with parameter θ ranging over $[0, x]$. More specifically, assuming that the prior distribution

$$\nu(M \in L_{\mathcal{B}[0,x]} | M \in \mathcal{B})$$

is a beta distribution with parameters α_0 and β_0 , that is

$$\nu(M \in L_{\mathcal{B}[0,x]} | M \in \mathcal{B}) = I_x(\alpha_0, \beta_0) \quad (10)$$

where I_x is defined as

$$I_x(\alpha, \beta) = \frac{B(x; \alpha, \beta)}{B(\alpha, \beta)} \quad (11)$$

where $B(x; \alpha, \beta)$ is the *regularized incomplete beta function*, defined as

$$B(x; \alpha, \beta) = \int_0^x \theta^{\alpha-1} \times (1 - \theta)^{\beta-1} \times d\theta \quad (12)$$

and $B(\alpha, \beta) = B(1; \alpha, \beta)$ is the beta function. Then

$$\text{CDF}_{\text{TV}(\mu)}(x) = I_x(n + \alpha_0, N - n + \beta_0) \quad (13)$$

where n is the number of positive pieces of evidence (corresponding to the cardinality of $\{i \in \mathcal{S} \mid (\mu i) = \text{True}\}$) and N is the total number of pieces of evidence (corresponding to the cardinality of $\mathcal{S}$). To justify that claim let us focus on

$$\begin{aligned} & \nu(M \in L_{\mathcal{B}[0,x]} | D_{\mathcal{S}}, M \in \mathcal{B}) \\ &= \frac{\prod_{i \in \mathcal{S}} \nu(D_i | M \in L_{\mathcal{B}[0,x]}) \times \nu(M \in L_{\mathcal{B}[0,x]} | M \in \mathcal{B})}{\prod_{i \in \mathcal{S}} \nu(D_i | M \in \mathcal{B})} \end{aligned} \quad (14)$$

$$= \frac{\prod_{i \in \mathcal{S}} \nu(D_i, M \in L_{\mathcal{B}[0,x]} | M \in \mathcal{B})}{\prod_{i \in \mathcal{S}} \nu(D_i | M \in \mathcal{B})} \quad (15)$$

$$= \frac{\prod_{i \in \mathcal{S}} \nu(D_i, M \in L_{\mathcal{B}[0,x]} | M \in \mathcal{B})}{\prod_{i \in \mathcal{S}} \nu(D_i, M \in L_{\mathcal{B}[0,1]} | M \in \mathcal{B})} \quad (16)$$

$$= \frac{\int_0^x \theta^{n+\alpha_0-1} \times (1 - \theta)^{N-n+\beta_0-1} \times d\theta}{\int_0^1 \theta^{n+\alpha_0-1} \times (1 - \theta)^{N-n+\beta_0-1} \times d\theta} \quad (17)$$

$$= \frac{B(x; n + \alpha_0, N - n + \beta_0)}{B(n + \alpha_0, N - n + \beta_0)} \quad (18)$$

which, by the definition of I_x , results in Equation 13. Let us detail this derivation. Equation 14 is a special case of Equation 8 where $L_{\mathcal{B}}$ is restricted to the class of Bernoulli processes where the parameter θ ranges from 0 to x . Equation 15 is a simplification of Equation 14 obtained from the fact that

$$\nu(D_i | M \in L_{\mathcal{B}[0,x]}) = \frac{\nu(D_i, M \in L_{\mathcal{B}[0,x]}, M \in \mathcal{B})}{\nu(M \in L_{\mathcal{B}[0,x]}, M \in \mathcal{B})} \quad (19)$$

$$= \frac{\nu(D_i, M \in L_{\mathcal{B}[0,x]} | M \in \mathcal{B})}{\nu(M \in L_{\mathcal{B}[0,x]} | M \in \mathcal{B})} \quad (20)$$

Equation 16 is merely obtain from the definition of conditional probability and the fact that $L_{\mathcal{B}[0,1]} = \mathcal{B}$. Equation 17 is derived as follows, considering first the numerator

$$\begin{aligned} & \prod_{i \in \mathcal{S}} \nu(D_i, M \in L_{\mathcal{B}[0,x]} | M \in \mathcal{B}) \\ &= \int_0^x \prod_{i \in \mathcal{S}} \nu(D_i, M \in L_{\mathcal{B}_{d\theta}} | M \in \mathcal{B}) \end{aligned} \quad (21)$$

$$= \int_0^x \prod_{i \in \mathcal{S}} \nu(D_i | M \in L_{\mathcal{B}_{d\theta}}) \times \nu(M \in L_{\mathcal{B}_{d\theta}} | M \in \mathcal{B}) \quad (22)$$

$$= \int_0^x \theta^n \times (1 - \theta)^{N-n} \times \nu(M \in L_{\mathcal{B}_{d\theta}} | M \in \mathcal{B}) \quad (23)$$

$$= \int_0^x \theta^n \times (1 - \theta)^{N-n} \times \frac{\theta^{\alpha_0-1} \times (1 - \theta)^{\beta_0-1} \times d\theta}{B(\alpha_0, \beta_0)} \quad (24)$$

$$= \frac{\int_0^x \theta^{n+\alpha_0-1} \times (1 - \theta)^{N-n+\beta_0-1} \times d\theta}{B(\alpha_0, \beta_0)} \quad (25)$$

Equation 21 is obtained by integrating over the parameter θ , using the additive law of probability, where $L_{\mathcal{B}_{d\theta}}$ is the set of Bernoulli processes with parameters set within an arbitrarily small interval centered around θ . Equation 22 is obtained from the definition of conditional probability. Equation 23 is obtained from the likelihood of $(D_i)_{i \in \mathcal{S}}$ given a Bernoulli process of parameter θ , where n is the number of positive pieces of evidence and N is the total number of pieces of evidence. Equation 24 comes from the beta distribution prior defined in Equation 10. Equation 25 is obtained by factorizing the powers over θ and $1 - \theta$. The denominator of Equation 17 is obtained analogously, then Equation 17 is obtained because $B(\alpha_0, \beta_0)$ on the numerator and denominator cancels out. Finally Equation 18 is obtained from the definition of the regularized incomplete beta function in Equation 12.

4.2 Implication Direct Introduction

This rule is not explicitly stated as such in the PLN book but can be derived from iteratively applying induction and revision, and reflects the formula of extensional inheritance/implication given in Section 2.4.1 *The Semantics of Inheritance* of the PLN book, at least the strength part. To obtain the confidence part we assume that the second order distribution is a beta distribution, like in the Section 4.2 *From Imprecise Probabilities to Indefinite Probabilities* of the PLN book. In order to derive a beta distribution as second order distribution we can assume an underlying Bernoulli process with parameter p with a beta distribution as prior. The Jeffreys prior where $\alpha = \beta = \frac{1}{2}$ is often the default choice. Given the PLN statement

$$P \Rightarrow Q$$

Let us express its truth value as the posterior probability of the parameter p of the underlying Bernoulli process. In order to map a Bernoulli process onto an implication we zoom-in to the data points where P is true.

NEXT

4.3 Priors

Typical priors for the beta distribution includes

- Haldane prior: $\alpha = \beta = 0$
- Jeffreys prior: $\alpha = \beta = \frac{1}{2}$
- Bayes prior: $\alpha = \beta = 1$

5 Other Inference Rules

5.1 Conjunction Introduction

Given $A \triangleq TV_A$ and $B \triangleq TV_B$ infer $A \wedge B \triangleq TV_{A \wedge B}$. For that we are going to describe the CDF of $TV_{A \wedge B}$ as it would be obtained by averaging over all possible combinations of A and B . Let us start with the discrete case, where

- n is the size of the universe,
- a is the size of A ,
- b is the size of B ,
- k is the size of $A \wedge B$,
- $n - a$ is the size of the complement of A ,
- $b - k$ is the size of $B \setminus A$.

Without loss of generality we can fix A and study the various ways the elements of B intersect with A such that $|A \wedge B| = k$

- C_k^a is the number of ways of placing the elements of $A \wedge B$ within A ,
- C_{b-k}^{n-a} is the number of ways of placing the elements of $B \setminus A$ within the complement of A ,
- C_b^n is the number of ways of placing the elements of B .

Thus the probability of $|A \wedge B| = k$

$$Pr(|A \wedge B| = k) = \frac{C_k^a \times C_{b-k}^{n-a}}{C_b^n}$$

Therefore the cumulative probability of $|A \wedge B| \leq k$ is

$$Pr(|A \wedge B| \leq k) = \sum_{i=0}^k \frac{C_i^a \times C_{b-i}^{n-a}}{C_b^n}$$

$$Pr(|A \wedge B| \leq k) = \sum_{i=0}^k \frac{\frac{a!}{i! \times (a-i)!} \times \frac{(n-a)!}{(b-i)! \times ((n-a)-(b-i))!}}{\frac{n!}{b! \times (n-b)!}}$$

$$Pr(|A \wedge B| \leq k) = \sum_{i=0}^k \frac{\frac{a!}{i! \times (a-i)!} \times \frac{(n-a)!}{(b-i)! \times (n-a-b+i)!}}{\frac{n!}{b! \times (n-b)!}}$$

Let us first think of the simpler case where $a = b$

$$Pr(|A \wedge B| \leq k) = \sum_{i=0}^k \frac{\frac{a!}{i! \times (a-i)!} \times \frac{(n-a)!}{(a-i)! \times (n-2 \times a + i)!}}{\frac{n!}{a! \times (n-a)!}}$$

$$Pr(|A \wedge B| \leq k) = \sum_{i=0}^k \frac{a!^2 \times (n-a)!^2}{i! \times (a-i)!^2 \times (n-2 \times a + i)! \times n!}$$

Let us simplify even further by assuming $a = \frac{n}{2}$

$$Pr(|A \wedge B| \leq k) = \sum_{i=0}^k \frac{\left(\frac{n}{2}\right)!^2 \times \left(\frac{n}{2}\right)!^2}{i! \times \left(\frac{n}{2} - i\right)!^2 \times \left(n - 2 \times \frac{n}{2} + i\right)! \times n!}$$

$$Pr(|A \wedge B| \leq k) = \sum_{i=0}^k \frac{\left(\frac{n}{2}\right)!^4}{i! \times \left(\frac{n}{2} - i\right)!^2 \times i! \times n!}$$

$$Pr(|A \wedge B| \leq k) = \sum_{i=0}^k \frac{\left(\frac{n}{2}\right)!^4}{i!^2 \times \left(\frac{n}{2} - i\right)!^2 \times n!}$$

Stirling approximation [DeepSeek Alert BEGIN]

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

First, approximate $\left(\frac{n}{2}\right)!$:

$$\left(\frac{n}{2}\right)! \approx \sqrt{2\pi \frac{n}{2}} \left(\frac{\frac{n}{2}}{e}\right)^{\frac{n}{2}}$$

Squaring both sides:

$$\left(\frac{n}{2}\right)!^2 \approx 2\pi \frac{n}{2} \cdot \left(\frac{n}{2e}\right)^n = \pi n \cdot \left(\frac{n}{2e}\right)^n$$

Next, approximate $n!$ using Stirling's formula:

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

Now, consider the ratio:

$$\frac{\left(\frac{n}{2}\right)!^2}{n!} \approx \frac{\pi n \cdot \left(\frac{n}{2e}\right)^n}{\sqrt{2\pi n} \left(\frac{n}{e}\right)^n}$$

Simplify the expression:

$$\begin{aligned} &= \frac{\pi n}{\sqrt{2\pi n}} \cdot \left(\frac{n}{2e}\right)^n / \left(\frac{n}{e}\right)^n \\ &= \sqrt{\frac{\pi n}{2}} \cdot \left(\frac{1}{2^n}\right) \end{aligned}$$

As n grows large, the term 2^n grows exponentially, while $\sqrt{\frac{\pi n}{2}}$ grows polynomially. Therefore, the ratio tends to zero as n approaches infinity. [DeepSeek Alert END]

Let us attempt to define the continuous case by studying the limit of the discrete case where n tends to $+\infty$. First let us reformulate the discrete case in terms of probabilities of A and B . Let us define the following probabilities

- $p_A = \frac{a}{n}$ is the probability of A , therefore $a = p_A \times n$,
- $p_B = \frac{b}{n}$ is the probability of B , therefore $b = p_B \times n$,
- $x = \frac{k}{n}$ is the threshold cumulative probability of $A \wedge B$, therefore $k = x \times n$,
- $p_{AB} = \frac{i}{n}$ is the probability of $A \wedge B$, therefore $i = p_{AB} \times n$,
- $p_{\bar{A}} = 1 - p_A = \frac{n-a}{n}$ is the probability of the complement of A , therefore $n - a = (1 - p_A) \times n$,
- $p_{B \setminus A} = p_B - p_{AB} = \frac{b-i}{n}$ is the probability of $B \setminus A$, therefore $b - i = (p_B - p_{AB}) \times n$.

Then the cumulative probability of $|A \wedge B| \leq k$ can be reformulated as

$$Pr\left(\frac{|A \wedge B|}{n} \leq x\right) = \sum_{i=0}^{x \times n} \frac{C_i^{p_A \times n} \times C_{p_B \times n - i}^{(1-p_A) \times n}}{C_{p_B \times n}^n}$$

$$\begin{aligned}
Pr(p_{AB} \leq x) &= \sum_{i=0}^{x \times n} \frac{C_i^{p_A \times n} \times C_{p_B \times n - i}^{(1-p_A) \times n}}{C_{p_B \times n}^n} \\
Pr(p_{AB} \leq x) &= \sum_{i=0}^{x \times n} \frac{\frac{(p_A \times n)!}{i! \times (p_A \times n - i)!} \times \frac{((1-p_A) \times n)!}{(p_B \times n - i)! \times (((1-p_A) \times n) - (p_B \times n - i))!}}{\frac{n!}{(p_B \times n)! \times (n - (p_B \times n))!}} \\
Pr(p_{AB} \leq x) &= \lim_{n \rightarrow +\infty} \sum_{i=0}^{x \times n} \frac{\frac{(p_A \times n)!}{i! \times (p_A \times n - i)!} \times \frac{((1-p_A) \times n)!}{(p_B \times n - i)! \times (((1-p_A) \times n) - (p_B \times n - i))!}}{\frac{n!}{(p_B \times n)! \times (n - (p_B \times n))!}} \\
Pr(p_{AB} \leq x) &= \\
\lim_{n \rightarrow +\infty} \sum_{i=0}^{x \times n} \frac{(p_A \times n)!}{i! \times (p_A \times n - i)!} \times \frac{((1-p_A) \times n)!}{(p_B \times n - i)! \times (((1-p_A) \times n) - (p_B \times n - i))!} \times \frac{(p_B \times n)! \times (n - (p_B \times n))!}{n!}
\end{aligned}$$

NEXT: use Stirling (or other see Bibliography/Mathematics) approximation to define a Riemman sum and turn it into integral <https://study.com/skill/learn/rewriting-the-limit-of-a-riemann-sum-as-a-definite-integral-explanation.html>

Let us redo the Stirling approximation with the full expression

$$\begin{aligned}
&Pr(p_{AB} \leq x) \\
&= \lim_{n \rightarrow +\infty} \sum_{i=0}^{nx} \frac{(np_A)!}{i!(np_A - i)!} \times \frac{(n(1-p_A))!}{(np_B - i)!(n(1-p_A) - np_B + i)!} \times \frac{(np_B)!(n(1-p_B))!}{n!}
\end{aligned}$$

Let us reform in 1 fraction

$$= \lim_{n \rightarrow +\infty} \sum_{i=0}^{nx} \frac{(np_A)!(n(1-p_A))!(np_B)!(n(1-p_B))!}{i!(np_A - i)!(np_B - i)!(n(1-p_A) - np_B + i)!n!}$$

[SPECIALIZATION BEGIN] What if $p_A = p_B = \frac{1}{2}$

$$= \lim_{n \rightarrow +\infty} \sum_{i=0}^{nx} \frac{\left(\frac{n}{2}\right)!^4}{i!^2 \left(\frac{n}{2} - i\right)!^2 n!}$$

[SPECIALIZATION END]

Let us use $t = \frac{i}{n}$ instead of i , thus $i = nt$, and attempt to introduce $\delta t = \frac{1}{n}$.

$$= \lim_{n \rightarrow +\infty} \sum_{t=0}^x \frac{(np_A)!(n(1-p_A))!(np_B)!(n(1-p_B))!}{(nt)!(np_A - nt)!(np_B - nt)!(n(1-p_A) - np_B + nt)!n!}$$

Let us factorize by n whenever possible.

$$= \lim_{n \rightarrow +\infty} \sum_{t=0}^x \frac{(np_A)!(n(1-p_A))!(np_B)!(n(1-p_B))!}{(nt)!(n(p_A - t))!(n(p_B - t))!(n(1-p_A - p_B + t))!n!}$$

Let us use Stirling approximation (not indicating limit and sum to save space).

$$\frac{\sqrt{2\pi n p_A} \left(\frac{n p_A}{e}\right)^{n p_A} \sqrt{2\pi(n(1-p_A))} \left(\frac{n(1-p_A)}{e}\right)^{n(1-p_A)} \sqrt{2\pi n p_B} \left(\frac{n p_B}{e}\right)^{n p_B} \sqrt{2\pi(n(1-p_B))} \left(\frac{n(1-p_B)}{e}\right)^{n(1-p_B)}}{\sqrt{2\pi n t} \left(\frac{n t}{e}\right)^{n t} \sqrt{2\pi(n(p_A-t))} \left(\frac{n(p_A-t)}{e}\right)^{n(p_A-t)} \sqrt{2\pi(n(p_B-t))} \left(\frac{n(p_B-t)}{e}\right)^{n(p_B-t)} \sqrt{2\pi(n(1-p_A-p_B+t))} \left(\frac{n(1-p_A-p_B+t)}{e}\right)^{n(1-p_A-p_B+t)} \sqrt{2\pi n} \left(\frac{n}{e}\right)^n}$$

Let us eliminate $\sqrt{2\pi n}$ in nominator and denominator (not indicating limit and sum to save space).

$$\frac{\sqrt{p_A} \left(\frac{n p_A}{e}\right)^{n p_A} \sqrt{1-p_A} \left(\frac{n(1-p_A)}{e}\right)^{n(1-p_A)} \sqrt{p_B} \left(\frac{n p_B}{e}\right)^{n p_B} \sqrt{1-p_B} \left(\frac{n(1-p_B)}{e}\right)^{n(1-p_B)}}{\sqrt{t} \left(\frac{n t}{e}\right)^{n t} \sqrt{p_A-t} \left(\frac{n(p_A-t)}{e}\right)^{n(p_A-t)} \sqrt{p_B-t} \left(\frac{n(p_B-t)}{e}\right)^{n(p_B-t)} \sqrt{1-p_A-p_B+t} \left(\frac{n(1-p_A-p_B+t)}{e}\right)^{n(1-p_A-p_B+t)} \sqrt{2\pi n} \left(\frac{n}{e}\right)^n}$$

On the numerator the exponents sum up to

$$n p_A + n(1-p_A) + n p_B + n(1-p_B) = n p_A + n - n p_A + n p_B + n - n p_B = 2n$$

On the denominator the exponents sum up to

$$n t + n(p_A-t) + n(p_B-t) + n(1-p_A-p_B+t) + n = n t + n p_A - n t + n p_B - n t + n - n p_A - n p_B + n t + n = 2n$$

Therefore e can be completely eliminated.

$$\frac{\sqrt{p_A} n p_A^{n p_A} \sqrt{1-p_A} n(1-p_A)^{n(1-p_A)} \sqrt{p_B} n p_B^{n p_B} \sqrt{1-p_B} n(1-p_B)^{n(1-p_B)}}{\sqrt{t} n t^{n t} \sqrt{p_A-t} n(p_A-t)^{n(p_A-t)} \sqrt{p_B-t} n(p_B-t)^{n(p_B-t)} \sqrt{1-p_A-p_B+t} n(1-p_A-p_B+t)^{n(1-p_A-p_B+t)} \sqrt{2\pi n} n^n}$$

There is a product of 4 n in the numerator and in the denominator, they can therefore be eliminated.

$$\frac{\sqrt{p_A} p_A^{n p_A} \sqrt{1-p_A} (1-p_A)^{n(1-p_A)} \sqrt{p_B} p_B^{n p_B} \sqrt{1-p_B} (1-p_B)^{n(1-p_B)}}{\sqrt{t} t^{n t} \sqrt{p_A-t} (p_A-t)^{n(p_A-t)} \sqrt{p_B-t} (p_B-t)^{n(p_B-t)} \sqrt{1-p_A-p_B+t} (1-p_A-p_B+t)^{n(1-p_A-p_B+t)} \sqrt{2\pi n} n^n}$$

Let us move the square roots in the exponents.

$$= \lim_{n \rightarrow +\infty} \sum_{t=0}^x \frac{p_A^{n p_A + \frac{1}{2}} (1-p_A)^{n(1-p_A) + \frac{1}{2}} p_B^{n p_B + \frac{1}{2}} (1-p_B)^{n(1-p_B) + \frac{1}{2}}}{t^{n t + \frac{1}{2}} (p_A-t)^{n(p_A-t) + \frac{1}{2}} (p_B-t)^{n(p_B-t) + \frac{1}{2}} (1-p_A-p_B+t)^{n(1-p_A-p_B+t) + \frac{1}{2}} \sqrt{2\pi n}^{n + \frac{1}{2}}}$$

[SPECIALIZATION BEGIN] What if $p_A = p_B = \frac{1}{2}$?

$$\begin{aligned} &= \lim_{n \rightarrow +\infty} \sum_{t=0}^x \frac{\left(\frac{1}{2}\right)^{2n+2}}{t^{2nt+1} \left(\frac{1}{2}-t\right)^{n-2nt+1} \sqrt{2\pi n}^{n+\frac{1}{2}}} \\ &= \lim_{n \rightarrow +\infty} \sum_{t=0}^x \frac{2^{n-2nt+1}}{2^{2n+2} \times t^{2nt+1} \times (1-2t)^{n-2nt+1} \times \sqrt{2\pi} \times n^{n+\frac{1}{2}}} \end{aligned}$$

[SPECIALIZATION END]

Let us exhibit δt by borrowing an n from $n^{\frac{1}{2}}$

$$= \lim_{n \rightarrow +\infty} \sum_{t=0}^x \frac{p_A^{np_A + \frac{1}{2}} (1-p_A)^{n(1-p_A) + \frac{1}{2}} p_B^{np_B + \frac{1}{2}} (1-p_B)^{n(1-p_B) + \frac{1}{2}}}{t^{nt + \frac{1}{2}} (p_A - t)^{n(p_A - t) + \frac{1}{2}} (p_B - t)^{n(p_B - t) + \frac{1}{2}} (1 - p_A - p_B + t)^{n(1-p_A-p_B+t) + \frac{1}{2}} \sqrt{2\pi n}^{n - \frac{1}{2}}} \delta t$$

Let us express exponents as sums

$$= \lim_{n \rightarrow +\infty} \sum_{t=0}^x \frac{p_A^{np_A + \frac{1}{2}} (1-p_A)^{n-np_A + \frac{1}{2}} p_B^{np_B + \frac{1}{2}} (1-p_B)^{n-np_B + \frac{1}{2}}}{t^{nt + \frac{1}{2}} (p_A - t)^{np_A - nt + \frac{1}{2}} (p_B - t)^{np_B - nt + \frac{1}{2}} (1 - p_A - p_B + t)^{n-np_A-np_B+nt + \frac{1}{2}} \sqrt{2\pi n}^{n - \frac{1}{2}}} \delta t$$

Let us turn exponent sums into products of exponents (removing limit, sum and moving component to the numerator or denominator so that its exponent is positive).

$$\frac{p_A^{np_A} p_A^{\frac{1}{2}} (1-p_A)^n (p_A - t)^{nt} (1-p_A-p_B+t)^{np_A} (1-p_A)^{\frac{1}{2}} p_B^{np_B} p_B^{\frac{1}{2}} (1-p_B)^n (p_B - t)^{nt} (1-p_A-p_B+t)^{np_B} (1-p_B)^{\frac{1}{2}}}{t^{nt} t^{\frac{1}{2}} (1-p_A)^{np_A} (p_A - t)^{np_A} (p_A - t)^{\frac{1}{2}} (1-p_B)^{np_B} (p_B - t)^{np_B} (p_B - t)^{\frac{1}{2}} (1-p_A-p_B+t)^n (1-p_A-p_B+t)^{nt} (1-p_A-p_B+t)^{\frac{1}{2}} \sqrt{2\pi n}^{n - \frac{1}{2}}} \delta t$$

Let us group coefficients by their exponents.

$$\left(\frac{p_A(1-p_A)p_B(1-p_B)^n}{t(p_A-t)(p_B-t)(1-p_A-p_B+t)} \right)^{\frac{1}{2}} \times \left(\frac{(1-p_A)(1-p_B)}{n(1-p_A-p_B+t)} \right)^n \times \left(\frac{p_A(1-p_A-p_B+t)}{(1-p_A)(p_A-t)} \right)^{np_A} \times \left(\frac{p_B(1-p_A-p_B+t)}{(1-p_B)(p_B-t)} \right)^{np_B} \times \left(\frac{(p_A-t)(p_B-t)}{t(1-p_A-p_B+t)} \right)^{nt} \times \frac{1}{\sqrt{2\pi}} \delta t$$

[DeepSeek ALERT BEGIN]

$$\int_0^x \frac{p_A^{p_A} (1-p_A)^{1-p_A} p_B^{p_B} (1-p_B)^{1-p_B}}{t^t (p_A - t)^{p_A - t} (p_B - t)^{p_B - t} (1 - p_A - p_B + t)^{1-p_A-p_B+t} \sqrt{2\pi}} dt$$

[DeepSeek ALERT END]

5.2 Implicant Conjunction Introduction

$$A \rightarrow C, B \rightarrow C \vdash A \wedge B \rightarrow C$$

Let us assume that

1. A and B are globally independent,
2. A and B are independent on C .

According Bayes' formula

$$P(C|A, B) = \frac{P(A, B|C) \times P(C)}{P(A, B)}$$

Because of assumption 1 and 2 we can derive

$$P(C|A, B) = \frac{P(A|C) \times P(B|C) \times P(C)}{P(A) \times P(B)}$$

Further more, according to Bayes' formula

$$P(A|C) = \frac{P(C|A) \times P(A)}{P(C)}$$

$$P(B|C) = \frac{P(C|B) \times P(B)}{P(C)}$$

Therefore

$$P(C|A, B) = \frac{\frac{P(C|A) \times P(A)}{P(C)} \times \frac{P(C|B) \times P(B)}{P(C)} \times P(C)}{P(A) \times P(B)}$$

which simplifies into

$$P(C|A, B) = \frac{P(C|A) \times P(C|B)}{P(C)}$$

6 conclusion

References